

RAJARAJAN M

Generative AI Engineer | LLM Applications | RAG Systems | LangChain | Docker

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PROFESSIONAL SUMMARY

Generative AI Engineer with hands-on experience building **production-ready LLM applications**, Retrieval-Augmented Generation (RAG) systems, and multimodal AI solutions. Delivered **3 end-to-end deployed AI systems** across legal, media, and multimodal domains. Skilled in **LangChain, FAISS, ChromaDB, HuggingFace, and Docker-based deployment**. Experienced in agentic AI workflows, prompt engineering, and scalable cloud deployment. Actively seeking **Generative AI / LLM Engineer roles (0-2 yrs)**.

TECHNICAL SKILLS

Programming:	Python, SQL, REST APIs, Git / GitHub, Environment Management
Generative AI & LLMs:	RAG Systems, LangChain, Agentic AI, Prompt Engineering, HuggingFace Transformers, GPT-2 Fine-Tuning, Multimodal AI, Gemini 2.5 Flash, LLM Orchestration
ML & NLP:	Transformers, Text Summarisation, Information Retrieval, Feature Engineering, Model Evaluation, Scikit-learn, Regression, Random Forest
Vector Databases:	FAISS, ChromaDB, Embeddings, Semantic Search, Document Chunking, Metadata Filtering
Deployment & DevOps:	Docker, Docker Hub, Streamlit, Gradio, python:3.12-slim containerisation, CI/CD pipelines, Scalable Deployment
Data Science:	Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Data Visualisation, Statistical Analysis

PROJECT EXPERIENCE

Legal Multi-Agent RAG System

Tech: Llama-3, LangChain, FAISS, ChromaDB, Gradio | github.com/RajaRajan71/Legal-_RAG_system

- Architected a **multi-agent RAG pipeline** for legal Q&A; using Llama-3, LangChain orchestration, and a citation validation agent
- Indexed and retrieved across **12+ legal documents** using FAISS vector store and ChromaDB with metadata filtering, achieving **sub-3s query response**
- Improved answer relevance by **~35%** through optimised chunking strategy and semantic metadata filtering vs. naive retrieval baseline
- Reduced LLM hallucinations by **~40%** by implementing a grounding-based citation validation layer before response generation
- Deployed interactive **Gradio web interface** enabling real-time legal querying with source citation display

Palludagam AI – Multimodal LLM System

Tech: Gemini 2.5 Flash, Python, Docker, Streamlit | github.com/RajaRajan71/palludagam-ai-hub

- Built production-ready multimodal AI app processing **3 input types** (image, audio, PDF) via Gemini 2.5 Flash API integration
- Reduced Docker image size by **~40%** using python:3.12-slim base, cutting cold-start time and cloud deployment costs
- Automated CI/CD pipeline via **Docker Hub** (rajann71/palludagam-ai-hub:v1), enabling one-command reproducible global deployment
- Designed real-time **Streamlit UI** with consistent cross-environment performance and zero "works on my machine" issues

GPT-2 Fine-Tuning & Deployment

Tech: HuggingFace Transformers, Tokenisation, Streamlit | github.com/RajaRajan71/final-project--guvi_llm

- Fine-tuned GPT-2 transformer model on a custom domain dataset achieving **stable convergence within 5 epochs** and **~30% improved** text generation quality

- Built custom tokeniser and preprocessing pipeline; optimised hyperparameters (learning rate, batch size, warmup steps) for best validation loss
- Deployed **production-ready text generation web app** on Streamlit with real-time inference and user-configurable generation parameters

AI-Powered Daily News Brief Generator

Tech: Transformers, NLP, API Integration, Streamlit | github.com/RajaRajan71/daily-news-brief-generator

- Built transformer-based summarisation system processing **50+ articles/day** across **5 topic categories** with personalised filtering
- Integrated **3 real-time external news APIs** to stream and process live data with automated deduplication and relevance ranking
- Deployed interactive Streamlit application with topic-preference UI, reducing reading time by an estimated **70%** per user session

ADDITIONAL PROJECTS (Classical ML & Data Science)

Singapore Resale Flat Price Prediction

Tech: Scikit-learn, Regression, EDA, Streamlit

github.com/RajaRajan71/Singapore-Resale-Flat-Prices-Predicting---project-5

- End-to-end regression ML pipeline; feature engineering, model validation, and Streamlit deployment. Achieved **R2 score of 0.91** on test set.

Industrial Copper Price & Sales Modelling

Tech: Random Forest, EDA

github.com/RajaRajan71/Industrial-copper-modeling

- Random Forest regression for industrial price forecasting; outlier treatment, skew correction and feature selection — improved RMSE by **~22%** vs baseline.

PhonePe Pulse Data Visualisation

Tech: Pandas, Data Visualisation

github.com/RajaRajan71/Phonepe-Pulse-Data-Visualization-and-Exploration-

- Analysed **1M+ transaction records**; EDA, trend analysis, and business intelligence dashboards for payment pattern insights.

EDUCATION

IIT-M GUVI — Advanced Programming Professional & Master Data Science

Chennai | Feb 2024 – Oct 2024

Specialisation: Large Language Models (LLMs), RAG Systems, MLOps, Generative AI, Deep Learning, Data Engineering, Python Programming

Paavai College of Engineering — B.E. Electronics & Communication Engineering

Namakkal | Aug 2019 – Apr 2024 | CGPA: 8.52 / 10

Relevant coursework: Data Structures & Algorithms, Python Programming, Signal Processing, Communication Systems, Microprocessors